Question #01

(A) & (B)

① Large (Alexnet) 11x11 convolution layer with stride = 4 followed by max-pooling is the input layer of the ALEXNET for downsampling the large input images.

② It is the VGG net which uses repeated 3x3 convo It has same effective receptive field as large filters with less parameters

③ Resnet => passing info from earlier layer to deeper layers by identity Skip / Shortcut connection.

④ Google-Lenet => Parallel convolution branches with varying Kernal sizes. (images objects can be of vastly different sizes)

⑤ Global Average Pooling => (NiN) => Replace massive parameter heavy layers with GAP right before classification layer.

③ Standard vggnet design Doubling number of feature maps while halving spatial resolution. (replace FC dense layers reduce parameter

(B)

(c)

	Alexnet	Vgg	GoogleNet	Resnet
Network Depth	8 layers Shallow	Deep 16-19	very Deep 22 layers	Extremely Deep 50,-152 in latest version
Conv filter Design	varied filters 3x3 5x5 11x11	Uniformly Small filters 3x3	multiscale & mixed	uniformly small 3x3 1x1 bottleneck.
Feature Extraction Strategy	Coarse to fine Spatial reduction using large initial Stride	Deep Stack of uniform Spatial Aggregations	Parallel multi Scale feature spatial Aggregations	Deep residual mapping with identity Shortcuts.
Information flow	Linear	Linear	Parallel multi path branching within blocks.	Non Sequential identity skip Connection
Computational Efficiency	Low efficiency heavy parameters	Poor efficiency high parameter counts	high efficiency	Very high Efficiency
Main Innovation	Cpu Trained deep CNN with relu & dropout	Standarized Vgg blocks 3x3	Split Transform merge (inception module)	skip Connections.

(d)

Plain Network :- Gradient must pass sequentially through every single weight matrix via chain rule. if weights are poorly initialized vanishing Gradient problem occurs. This makes earliest layers learn very slow or stop learning altogether.

Shortcut Network :- identity shortcut allow gradients to flow directly from deep layers to shallow layers without being altered by weight multiplication $\frac{\partial f_n}{\partial x}$ the 1 ensures gradient flow

flow back to early layers unimpeded. prevent gradient vanishing regardless depth.

(E)

(i) reduce parameters count $7 \times 7 \times C^2 = 49C^2$ and $3(3 \times 3) = 27C^2$ reduction of roughly 45% in parameters low memory usage.

(ii) 3X3 apply non linear activation function three times instead of 1. Enable to learn far more complex features

if 3X3 convolution layers only used it would lose core property of multi-scale feature Extraction

(1) Scale Diversification (2) Dimensionality Control

(F)

AlexNet ⟹ Vgg ⇒ GoogleNet ⇒ ResNet

AlexNet ⇒ introduced GPUs, relu, DropOut.

Vgg ⇒ homogenous 3x3 network used ad-hoc filter sizes (11x11, 5x5)

GoogleNet ⇒ Vgg had computation, memory high it introduce inception module 1x1 bottleneck convolutions.

Resnet ⇒ introduced skip connection solve deep layer training

Question #4

(A)

$(11 \times 11 * 3 * 1) \times 96 = (363 + 1) \times 96 = 34\,944$

$(5 \times 5 \times 96 + 1) \times 256 = (2400 + 1) \times 256 = 614\,656$

$(3 \times 3 \times 256 + 1) \times 384 = (2304 + 1) \times 384 = 885\,120$

$(3 \times 3 \times 384 + 1) \times 384 = (3456 + 1) \times 384 = 1\,327\,488$

$(3 \times 3 \times 384 + 1) \times 256 = (3456 + 1) \times 256 = 884\,992$

(B)

(1) $227 \times 227 \times 3$

$\frac{227 - 11}{4} + 1 = 54 + 1 = 55 \times 55 \times 96$

maxpool (3x3 stride 2)

$\frac{55 - 3}{2} + 1 = 26 + 1 = 27 \times 27 \times 96$

Convo (5x5 pad 2)

$\frac{27 - 5 + 4}{1} + 1 = 27 \times 27 \times 256$

maxpool (3x3 stride 1)

$= \frac{27 - 3}{2} + 1 = 13 \times 13 \times 256$

Convo3 (3x3 pad 1)

$\frac{13 - 3 + 2}{1} + 1 = 12 + 1 = 13 \times 13 \times 384$

Convo4 (3x3 pad 1)

$\frac{13 - 3 + 2}{1} + 1 = 13 \times 13 \times 384$

Convo 5 (3x3 pad 1)

$\frac{13 - 3 + 2}{1} + 1 = 13 \times 13 \times 256$

maxpool (3x3, S=2)

$\frac{13 - 3}{2} + 1 = 6 \times 6 \times 256$

input size of FC1

$6 \times 6 \times 256 = 9216$

(C)

FC1 (9216 → 4096)

$= (9216+1) \times 4096 = 9217 \times 4096 = 37752832$

FC2 (4096 → 4096)

$(4096+1) \times 4096 = 4097 \times 4096 = 16781312$

FC3 (4096 → 1000)

$4097 \times 1000 = 4097000$

(D)

$39744 + 614656 + 885120 + 1327488 + 884992 = 3847200$

Convolution Sum $= 3847200$

FC sum $= 37752832 + 16781312 + 4097000$

$= 58631144$

total parameters $= 3847200 + 58631144 = 62378344$

Question #3

(1)

1. Convolution layer 1 (width × height × channels + 1) × No. of filters

$= (5 \times 5 + 1 \times 1) \times 6 = 26 \times 6 = 156$

2. convolution layer 2

$(5 \times 5 \times 6 + 1) \times 16 = 151 \times 16 = 2416$ parameters

(2)

1. FC Dense layer

$(400 + 1) \times 120 = 401 \times 120 = 48120$ parameters

2. FC2 layer

(3) $(120 + 1) \times 84 = 121 \times 84 = 10164$

1. Final output layer

$85 \times 10 = 850$ Parameters

(4)

$$156 + 2416 + 48120 + 10164 + 850$$

$$= 601706 \text{ parameters.}$$

CamScanner